The Sugar Spinner: Engineering a Plug-and-Play Biological System for In-Situ Fibre Synthesis

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1 Introduction

Levan is a naturally occurring fructan present in many plants and microorganisms. This polymer is made up of fructose, a monosaccharide sugar, primarily connected by $\beta(2 \rightarrow 6)$ glycosidic linkages. Levan chains are generally characterised as either low molecular weight (LMW) or high molecular weight (HMW) polymers, both of which are considered to be dietary fibre but which can exhibit different structural and behavioural properties [15, 14, 12].

Levan polymer chains can be formed through the activity of the enzyme levansucrase. Levansucrase is an enzyme that catalyzes the chemical reaction of sucrose into levan, however it also catalyzes the hydrolysis of sucrose and has levanase side-activity that can depolymerise the very levan polymer chains that it also creates [11]. The result is a complex and paradoxical system that can form varied numbers and weights of levan polymer chains [6].

In general, a system containing the levansucrase enzyme and sucrose can result in any of the following reactions:

1. Sucrose cleavage: the glycosidic bond is broken, converting sucrose into glucose and a fructose-enzyme complex
2. Transfructosylation: a fructose unit from a donor (sucrose) is transferred to an acceptor (either sucrose or a free fructose) to produce prebiotic fructooligosaccharides (FOS) (short chain levans)
3. Levan chain polymerisation: a fructose unit is transferred from sucrose to a growing acceptor chain
4. Levan chain depolymerisation/exo-hydrolysis: levan chains are broken down by cleaving subunits specifically from the ends of the chain
5. Hydrolysis: a water molecule is the acceptor of the fructose-enzyme complex, breaking apart the complex and releasing free fructose

The use of levansucrase to convert sucrose to dietary fibre in the context of beverage production is not a new concept and has good potential [2].

2 Identifying and evaluating levansucrase variants

Levansucrase can be synthesised by a variety of different organisms and different variants can have widely different properties regarding its ability to synthesis levan polymer chains of different molecular weights [7, 9, 8].

2.1 *Bacillus subtilis*

An obvious choice of organism from which to extract the levansucrase enzyme is *Bacillus subtilis*, partly due to the extensive studies that have been carried out [1].

2.2 *Zymomonas mobilis*

A potential alternative to *Bacillus subtilis* is the bacterium *Zymomonas mobilis*. It is already used in food and beverage production, has GRAS (Generally Recognised as Safe) status and has been identified as having untapped potential in the food and beverage industry [10].

One of the major advantages of *Zymomonas mobilis* over *Bacillus subtilis* is its stability in low pH environments [13, 5]. Wildtype enzymes have optimum pH of 5 with many variants having stability down to a pH of 3.7 and thermal stability up to 60°C. In particular, we identify the S345A-LSZm variant as having good levels of activity whilst being particularly robust to stressful environments, although there is evidence to suggest that while this variant has good overall activity, it struggles to synthesis High Polymer Weight (HMW) levan chains [13].

Some evidence to suggest that levansucrase from *Zymomonas mobilis* does not produce the long chain/high molecular weight polymers however some literature suggests that it does create the HMW polymers [5]. [16]

2.3 Shifting the optimal pH

Levansucrase belongs to the 68 glycoside hydrolase family (GH68). The enzyme contains a central catalytic pocket in which three conserved residues are essential for catalysis: an aspartic acid (Asp) acting as the nucleophile, a glutamic acid (Glu) serving as the general acid, and a second aspartic acid involved in stabilizing the transition state.

According to the canonical double-displacement mechanism of retaining glycosidases, the nucleophile (Asp) must be deprotonated to perform nucleophilic attack, while the general acid (Glu) needs to be protonated during substrate binding to facilitate glycosidic bond cleavage. Thus, the catalytic activity requires these residues to adopt distinct protonation states.[4]

However, the intrinsic pKa of the Glu side chain is approximately 4.15, which would suggest that it is predominantly deprotonated at pH 6, close to the reported optimal pH of levansucrase. This apparent discrepancy indicates that the effective pKa of catalytic residues in the enzyme active site is significantly modulated by the local microenvironment. Indeed, Meng *et al.*[3] proposed that this arises from the cumulative effects of surrounding residues and their interactions with Glu.

Therefore, we hypothesize that mutations in residues interacting with the catalytic Asp and Glu could alter their effective pKa values by reshaping the local electrostatic and hydrogen-bonding environment, thereby shifting the optimal pH of the enzyme. In support of this idea, the mutation of a serine(Ser) residue that interacts with nucleophilic Asp to alanine(Ala) was shown to reduce the pH stability range of the enzyme [13], highlighting the importance of local residue interactions in maintaining catalytic function.

Together, rational engineering of residues surrounding the catalytic site represents a promising strategy to decrease the optimal pH of levansucrase.

To identify potential mutation targets in *bacillus subtilis*, we examined the three-dimensional structure of levansucrase using available PDB data.

Residues that are likely to interact with the catalytic amino acids were selected, as they may influence the local microenvironment and, consequently, the effective pKa of the catalytic residues.

- Asp86 (nucleophile): Ser164, Leu109, Ser412.
- Asp247: Arg343, Thr180, Arg246, Ala344, Ser164.
- Glu342 (general acid): Arg343, Thr357, Asp358, Tyr385, Arg360, Glu340, Ser359, Arg246, Asp247.

These residues are positioned within the catalytic pocket and may contribute to electrostatic interactions or hydrogen-bonding networks.

Therefore, mutations at these positions could potentially alter the protonation states of the catalytic residues and shift the optimal pH of levansucrase.

Based on this structural analysis, these residues are proposed as candidates for further mutagenesis.

3 Computational modelling of levansucrase kinetics

3.1 Elementary reaction scheme

$$
\begin{aligned}
S + LS &\xrightarrow{k_1} G + F\text{-}LS && \text{(sucrose cleavage)}\\
F\text{-}LS + H_2O &\xrightarrow{k_2} F + LS && \text{(hydrolysis)}\\
F\text{-}LS + S &\xrightarrow{k_3} L_1 + LS && \text{(transfructosylation with sucrose acceptor)}\\
F\text{-}LS + F &\xrightarrow{k_3} L_1 + LS && \text{(transfructosylation with fructose acceptor)}\\
F\text{-}LS + L_n &\underset{k_5}{\overset{k_4}{\rightleftharpoons}} L_{n+1} + LS && \text{(polymerisation/depolymerisation)}\\
F\text{-}LS + L_N &\xrightarrow{k_4} L_{\text{long}} + LS && \text{(polymerisation into absorbing state)}
\end{aligned}
\tag{1}
$$

where $n \in \{1, \ldots, N-1\}$ and
LS: Levansucrase
S: Sucrose
G: Glucose
$F\text{-}LS$: Fructose-Levansucrase complex
L_n: Levan chain of length n
L_{long}: High Polymner Weight levan chain that can no longer be degraded

3.2 Mass-action ODE model

$$\frac{dS(t)}{dt} = -k_1\, LS(t)\, S(t) - k_3\, F\text{-}LS(t)\, S(t) \tag{2}$$

$$\frac{dLS(t)}{dt} = k_2\, F\text{-}LS(t) - k_1\, LS(t)\, S(t) + k_3\, F\text{-}LS(t)\, S(t) + k_3\, F\text{-}LS(t)\, F(t) \tag{3}$$

$$+ k_4\, F\text{-}LS(t)\, L2(t) + k_4\, F\text{-}LS(t)\, L_1(t) - k_5\, LS(t)\, L_2(t) - k_5\, LS(t)\, L_3(t) \tag{4}$$

$$\frac{dG(t)}{dt} = k_1\, LS(t)\, S(t) \tag{5}$$

$$\frac{dF\text{-}LS(t)}{dt} = -k_2\, F\text{-}LS(t) + k_1\, LS(t)\, S(t) - k_3\, F\text{-}LS(t)\, S(t) - k_3\, F\text{-}LS(t)\, F(t) \tag{6}$$

$$- k_4\, F\text{-}LS(t)\, L2(t) - k_4\, F\text{-}LS(t)\, L_1(t) + k_5\, LS(t)\, L_2(t) + k_5\, LS(t)\, L_3(t) \tag{7}$$

$$\frac{dF(t)}{dt} = -k_2\, F\text{-}LS(t) - k_3\, F\text{-}LS(t)\, F(t) \tag{8}$$

$$\frac{dL_1(t)}{dt} = k_3\, F\text{-}LS(t)\, S(t) + k_3\, F\text{-}LS(t)\, F(t) - k_4\, F\text{-}LS(t)\, L_1(t) + k_5\, LS(t)\, L_2(t) \tag{9}$$

$$\frac{dL_n(t)}{dt} = k_4 F\text{-}LS(t) L_{n-1} + k_5\, LS(t) L_{n+1} - k_5\, LS(t) L_n - k_4\, F\text{-}LS(t) L_n(t) \tag{10}$$

$$\frac{dL_N(t)}{dt} = k_4\, F\text{-}LS(t) L_{N-1} - k_5\, LS(t) L_N(t) - k_4\, F\text{-}LS(t) L_N \tag{11}$$

$$\frac{dL_{long}(t)}{dt} = k_4\, F\text{-}LS(t) L_N(t) \tag{12}$$

3.3 Michaelis-Menten ODE model

The enzyme levansucrase (LS) cycles between its free form and a fructosyl-enzyme intermediate (F-LS) via a ping-pong mechanism. The conservation law is

$$[LS] + [F\text{-}LS] = E_{\text{total}}. \tag{13}$$

Table 1: Reaction network. S = sucrose, G = glucose, F = fructose, L_n = levan chain of DP n, B = branch points.

#	Reaction	Rate expression	Description
1	$S + LS \to G + F\text{-}LS$	$v_1 = \dfrac{k_{\text{suc}} \cdot [LS] \cdot [S]}{K_m^S + [S]}$	Sucrose cleavage
2	$S + F\text{-}LS \to L_1 + LS$	$v_2 = \dfrac{k_{\text{init}}^S \cdot [F\text{-}LS] \cdot [S]}{K_m^{\text{init},S} + [S]}$	Initiation (sucrose acceptor)
3	$F + F\text{-}LS \to L_1 + LS$	$v_3 = \dfrac{k_{\text{init}}^F \cdot [F\text{-}LS] \cdot [F]}{K_m^{\text{init},F} + [F]}$	Initiation (fructose acceptor)
4a	$F\text{-}LS + L_n \to L_{n+1} + LS$	$v_4 = \dfrac{k_{\text{elong}} \cdot [F\text{-}LS] \cdot \mu_0}{K_m^{\text{elong}} + \mu_0}$	Elongation (β-2,6)
4b	$F\text{-}LS + L_n \to L_n^{(\text{br})} + LS$	$v_{4b} = \dfrac{k_{\text{br}} \cdot f_{\text{DP}} \cdot f_{\text{sat}} \cdot [F\text{-}LS] \cdot (\mu_1 - \mu_0)}{K_m^{\text{br}} + (\mu_1 - \mu_0)}$	Branching (β-2,1)
5	$L_n + LS \to L_{n-1} + F\text{-}LS$	$v_5 = \dfrac{k_{\text{depol}} \cdot [LS] \cdot \mu_0}{K_m^{\text{depol}} + \mu_0}$	Depolymerisation
6	$F\text{-}LS \to LS + F$	$v_6 = k_{\text{hyd}} \cdot [F\text{-}LS]$	Hydrolysis

Branching regulators. Branching is modulated by a sigmoid degree of polymerisation (DP) activation and a saturation cap:

$$f_{\text{DP}} = \frac{1}{1 + e^{-(\bar{n} - n_{\text{thresh}})}}, \qquad f_{\text{sat}} = \max\left(0,\ 1 - \frac{B}{\mu_1 \cdot \beta_{\max}}\right), \tag{14}$$

where $\bar{n} = \mu_1/\mu_0$ is the number-average degree of polymerisation.

The state vector is $\mathbf{y} = [S,\ G,\ F,\ F\text{-}LS,\ \mu_0,\ \mu_1,\ \mu_2,\ B]$ (8 states).

Small molecules.

$$\frac{d[S]}{dt} = -v_1 - v_2 \tag{15}$$

$$\frac{d[G]}{dt} = +v_1 \tag{16}$$

$$\frac{d[F]}{dt} = +v_6 - v_3 \tag{17}$$

$$\frac{d[F\text{-}LS]}{dt} = +v_1 + v_5 - v_2 - v_3 - v_4 - v_{4b} - v_6 \tag{18}$$

Levan chain population (moment method). The k-th moment of the chain length distribution is $\mu_k = \sum_i c_i\, n_i^k$, where c_i is the molar concentration of chains with DP n_i.

$$\frac{d\mu_0}{dt} = +(v_2 + v_3) \tag{19}$$

$$\frac{d\mu_1}{dt} = +v_4 + v_{4b} + 2(v_2 + v_3) - v_5 \tag{20}$$

$$\frac{d\mu_2}{dt} = (2\bar{n} + 1)(v_4 + v_{4b}) + 4(v_2 + v_3) - (2\bar{n} - 1)\, v_5 \tag{21}$$

The second moment equation uses the mean-field closure approximation $n \approx \bar{n}$.

Branch points.

$$\frac{d[B]}{dt} = +v_{4b} \tag{22}$$

$$\text{Fructose units:} \quad [S](t) + [F](t) + [F\text{-}LS](t) + \mu_1(t) = S_0 \tag{23}$$

$$\text{Enzyme:} \quad [LS](t) + [F\text{-}LS](t) = E_{\text{total}} \tag{24}$$

$$\bar{n} = \frac{\mu_1}{\mu_0} \qquad \text{Number-average DP} \tag{25}$$

$$\text{PDI} = \frac{\mu_2\, \mu_0}{\mu_1^2} \qquad \text{Polydispersity index} \tag{26}$$

$$Y = \frac{\mu_1}{S_0} \qquad \text{Levan yield (fructose basis)} \tag{27}$$

$$\phi_{\text{br}} = \frac{B}{\mu_1} \qquad \text{Branch fraction } (\beta\text{-2,1 linkages}) \tag{28}$$

$$\frac{[G]}{[F]} \qquad \text{Glucose/fructose ratio } (> 1 \text{ indicates net polymerisation}) \tag{29}$$

3.4 Agent Based spatial model

We also developed an agent based spatial model that considers individual molecules and reactions.

To do so, we define a grid in which the different species can diffuse and probabilistically react according to our reaction scheme and rates. This allows us to consider the effects of space, volume and size on the reactions, whereas the ODE models cannot consider these factors.

One of the major benefits to the agent based model is that it gives us the ability to map the densities of the different species. We are able to track the distribution of different length levan chains to better analyse the effect of different parameters not only on overall levan production but on the production of the different forms of levan.

The major drawback to a model such as this is the computational expense - hence the benefit of having both ODE models (which are fast but less realistic) and the agent based model.

Table 2: Kinetic parameters. Default values for *B. subtilis* levansucrase at 37°C, pH 6.

Symbol	Units	Default	Description
E_{total}	M	10^{-6}	Total enzyme concentration
k_{suc}	s^{-1}	150	Sucrose cleavage turnover number
K_m^S	M	0.025	Michaelis constant for sucrose (cleavage)
k_{hyd}	s^{-1}	30	Hydrolysis rate constant (first-order)
k_{init}^S	s^{-1}	20	Initiation turnover (sucrose acceptor)
$K_m^{\text{init},S}$	M	0.05	K_m for sucrose as initiation acceptor
k_{init}^F	s^{-1}	5	Initiation turnover (fructose acceptor)
$K_m^{\text{init},F}$	M	0.10	K_m for fructose as initiation acceptor
k_{elong}	s^{-1}	80	Elongation turnover (β-2,6)
K_m^{elong}	M	0.01	K_m for chain non-reducing end
k_{br}	s^{-1}	5	Branching turnover (β-2,1)
K_m^{br}	M	0.02	K_m for internal fructose units
k_{depol}	s^{-1}	2	Depolymerisation turnover
K_m^{depol}	M	0.005	K_m for chain binding (depolymerisation)
n_{thresh}	–	5	DP threshold for branching activation
$\beta_{\max}$	–	0.15	Maximum branch linkage fraction
T	K	310.15	Temperature
pH	–	6.0	pH
E_a	J/mol	45000	Activation energy (Arrhenius)
pH_{opt}	–	6.0	Optimal pH
σ_{pH}	–	1.5	pH activity bell-curve width

4 Future prospects

In future, the models provided in this project can be used with a wide parameter scan and sensitivity analysis to assess the effect of each of the parameters on the model. This will ultimately provide insight on which parameters have the greatest effect on levan yield, helping to identify potential engineering targets.

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